SYNOPTIX
systems thinking

SYNOPTIX RESPONSIBLE AI HACKATHON

WELCOME PRESENTATION

24th-25th of October

COMMERCIAL IN CONFIDENCE

SYNOPTIX

HOUSEKEEPING

SYNOPTIX
systems thinking

Housekeeping

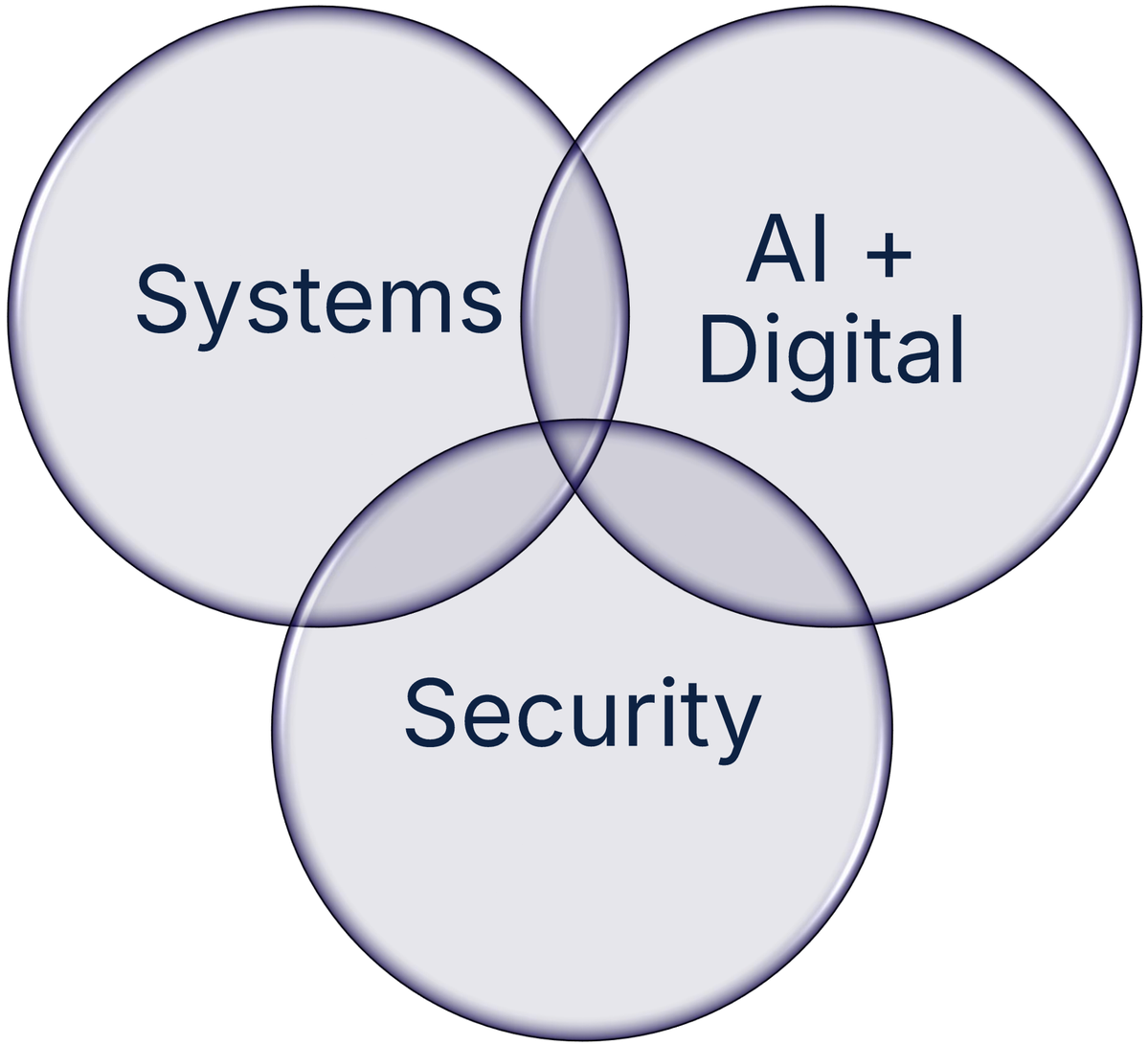
- Toilets
- Fire exits
- Building Access
- Lunch
- Photography consent

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INTRODUCTION TO SYNOPTIX

SYNOPTIX
systems thinking

Who are Synoptix?



SYNOPTIX
systems thinking

Partners and Collaborators

Industry

BMT

LEONARDO

BAE SYSTEMS

South East London Integrated Care System

NetworkRail

verview

ONE BEYOND

smiths detection

ROLLS ROYCE

amberjack

ROYAL NAVY

GE VERNOVA

ARX ROBOTICS

Academia

KING'S College LONDON

UNIVERSITY of York

UNIVERSITY OF LEICESTER

University of Exeter

UWE Bristol University of the West of England

University of BRISTOL

UNIVERSITY OF BATH

University of Southampton

Loughborough University

UNIVERSITY OF PLYMOUTH

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UWE Bristol

HackathonsUK

Institution of MECHANICAL ENGINEERS

UWE COMPUTER SCIENCE SOCIETY UNIVERSITY OF THE WEST OF ENGLAND BRISTOL

IEEE

IEEE Branch at UWE





Meet the Team

Chair IEEE Students Branch – Ruth

Vice Chair – Jack

Secretary – Archie

Treasurer - Sasirindu

Social Media Coordinator – Anton

Events Coordinator – Isaac





IEEE

United Kingdom and Ireland Section

What is IEEE?

- IEEE = *Institute of Electrical and Electronics Engineers*
- World’s largest technical professional organisation
- 400,000+ members in 160+ countries
- Supports innovation in engineering, computing, and technology
- Publishes leading research and journals
- Develops global technical standards
- Hosts conferences, competitions, and workshops
- Provides a network of professionals, researchers, and students

IEEE
United Kingdom and Ireland Section

Institution of MECHANICAL ENGINEERS

WESTERN REGION

Mission

*The IMechE Western Region will...support and grow **our committees...**have full **regional impact** through a range of **innovative activities**. We will ...represent our **regional Member's** interests throughout.*

Impact

STEM
Industry Engagement
Career Support

Membership Development

Learning and Development - Lectures, visits and events.
Networks
Professional development support



Region covers 6 areas:

Bath & Bristol
Devon & Cornwall
Devon & Somerset
Gloucester
Swindon & North Wiltshire Area
Channel Islands

Improving the world through engineering

BATH AND BRISTOL AREA

Mission

*Enhance the experience and engagement of **local members** with the institute through **engineering-based activities** and **events.***

Strategies

- Evergreen the committee
- Define interests of local membership
- Create a calendar of engaging events

B&B committee



Dr Owen Lewis

Chair

Dr Zeyad Al-Shibaany

Vice Chair, Treasurer





Kofi Fox

Secretary

Institution of MECHANICAL ENGINEERS

JOIN THE B&B COMMITTEE

Have an idea for an event in the Bath and Bristol area?

Join us and help make it happen

Strategies
- Evergreen the committee

Commitment: *1-2hrs per month*
Opportunities: *Rotation of positions encouraged*
Duration: 12-24 month service

Open roles:
- Events coordinator
- Communications and Social media representative
- Anyone who wants to contribute

Improving the world through engineering

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INTRODUCTION TO HACKATHON

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AI is showing up in more of daily life every year

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AI is showing up in more of daily life every year

Rapid growth has brought a responsibility gap

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Challenge

- **Your challenge:** Build something that helps close the responsibility gap. We want ideas that make AI systems more trustworthy; whether that means fairer, more transparent, more private, more sustainable, or better governed.

- **The goal:** Empower developers and organisations to adopt AI that is trustworthy and fair, so AI benefits everyone.

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Event Structure

Responsible AI is the **one overall challenge** but there are **five tracks**. You can:

- Pick **one track** and go deep,
- Build something that touches **two or more tracks**, or
- If you already have an idea, go **off the tracks**.

You do not need to cover every track. Judges are looking for depth and rigour on the track(s) you choose, not breadth across all five.

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Event Tracks

ID	Track	Description
1	Bias Mitigation	Detect or reduce unfair treatment in AI models and data
2	Explainability & Transparency	Make AI decisions understandable to the people affected by them
3	Privacy & Security	Protect user data without wrecking AI performance
4	Sustainability & Accessibility	Make AI systems cheaper to run and usable by more people
5	Governance & Compliance	Monitor and enforce ethical AI standards in practice

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Presentations

The main hackathon space will be turned into an exhibition style venue

- Teams will present at their tables
- Please keep a **minimum of 2** at the table at any time
- Judges will be walking around assessing

Judges



Keith Speed

Business Development Director, Synoptix



Sam Farrow

Technology Director, Synoptix



Ghaly Mikhail

Mechatronics and Automotive Lecturer, UWE



Zeyad Al-Shibaany

Associate Professor in Digital Manufacturing, UWE

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Schedule – Day 1

	Main Event Space	Lecture Theatre	Side Room
9:00	9:00-9:30 Volunteers to Arrive 9:30-10:00 Participants to Arrive	9:30-9:45 Volunteer briefing	
10:00		10:00-11:00 Opening Ceremony	
11:00	11:00-17:00 Main Hackathon Event	11:30-12:00 Presentation 1	
12:00			12:00-13:00 Lunch
13:00		13:00-13:30 Chess Tournament Setup	
14:00		13:30-16:30 Chess Tournament	
15:00	15:00-16:00 Careers Fair		
16:00	11:00-17:00 Main Hackathon Event		
17:00	Event Close	Event Close	Event Close

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Schedule – Day 2

	Main Event Space	Lecture Theatre	Side Room
8:00	8:30-8:45 Volunteers to Arrive	8:45-9:00 Volunteer Briefing	
9:00	9:00-11:00 Main Hackathon Event		
10:00			10:30-11:00 Lunch Setup
11:00		11:30-12:00 Judging Briefing	11:00-12:00 Networking Lunch
12:00	12:00-14:00 Hackathon Presentations		
13:00			
14:00			14:00-15:00 Judge Deliberations
15:00		15:00-15:30 Awards Ceremony	
16:00	Event Close	Event Close	Event Close

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INTRO TO AI ASSURANCE

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Three things:

- What did you think AI Assurance was?
- Examples of poor assurance
- Our challenge to you all

Your definitions...



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Some observations:

- Shared vocabulary issue: from testing outputs and monitoring models, to following regulations, using AI responsibly, protecting businesses and communicating evidence
- Balance between increasing trust and knowing when not to trust
- “Proving safety” versus acknowledging uncertainty
- Mixed “objects of assurance”
- Human oversight was often mentioned – but much less detail on how to actually achieve it
- Assurance and responsible use are different things
- Assurance is more than a collection of tests

Five key examples:



St George's
Bias mitigation



Michigan MiDAS
Explainability & Transparency



Roomba
Privacy & Security



Chilean data centre
Sustainability & Accessibility



Air Canada
Accountability & Compliance

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Our challenge to you:

- Consider the trustworthiness of what you intend to build: how do we measure, evaluate, communicate, prove, justify, argue that what we are doing should have trust placed in it
- Take a broad approach to coming up with an idea – a simple idea, executed very well, could be very compelling
- Consider technologies beyond LLMs – e.g. many predictive algorithms are much simpler classifier models – but can still cause a great deal of harm
- Talk about humans, how they work, how they think, how they interact: the user is part of the system, not external to it
- Think about external factors: funding, politics, scaling, adoption – we don’t build in bubbles.

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Contact Us


530 Bristol Business Park, BS16 1EJ


+44(0) 117 943 0707


info@synoptix.co.uk


www.synoptix.co.uk